REDROB · DATA & AI CHALLENGE

HireFit Ranker

Ranks careers, not keywords.

A deterministic, offline, CPU-only engine that reads who actually fits a role — then returns a 100-candidate shortlist a recruiter can trust.

100K	**184s**	**0.894**	**0**
candidates scored	CPU runtime (≤300s)	NDCG@10 *	honeypots in top-100

* *NDCG@10 on an independent LLM-judged sample (dev proxy) — see slide 8. Not the hidden challenge score.*

THE PROBLEM

Keyword filters miss the right people — and rank the wrong ones first.

The dataset is engineered to punish keyword matching. Four trap types sink a naive ranker.

Keyword stuffers

Wrong-role profiles padded with AI skills — a Marketing Manager listing 9 AI keywords.

Wrong-role profiles

Perfect skill lists attached to titles that never shipped a ranking system.

Honeypots

Subtly impossible profiles — “expert” in 10 skills with 0 months used. Forced to tier 0.

Unavailable-but-perfect

Great on paper, but 5% recruiter response and not logged in for months.

The real task: rank candidates the way a great recruiter would — reading career history and behavior, not counting keywords. A candidate who *built* a recommender at a product company outranks one who merely *lists* “RAG” and “Pinecone.”

THE INSIGHT

The JD lists keywords; the role is hiring for evidence.

Our whole design follows from reading the gap between what the JD says and what it means.

What the JD says

“5–9 years, embeddings, retrieval, ranking, NLP, deep learning, PyTorch, TensorFlow…”

embeddings

retrieval

ranking

NLP

5–9 yrs

The trap: skill-keyword overlap is high for stuffers and low for the engineers who actually shipped the systems.

What the role means

- **Shipped work** — Built & deployed ranking / search / recsys in production.
- **Career trajectory** — Consistent growth in relevant, product-company roles.
- **Right seniority** — **5–9 years of judgment — penalize overshoot and juniors.**
- **Actually available** — Responsive, open to work, reachable notice period.

THE SYSTEM

One offline pipeline scores every candidate — no LLM in the ranking loop.

Six deterministic stages turn a raw profile into a trusted rank. Runs on CPU, no network, no GPU.

1. **Parse** — structured text
2. **BM25** — lexical relevance
3. **Semantic** — concept expansion
4. **28 features** — recruiter matrix
5. **Base score** — weighted aggregate
6. **Guardrails** — behavior · honeypot · disqualifier

Final score = base score × behavioral × honeypot × disqualifier

Guardrails are multiplicative — a strong-on-paper profile that is impossible, off-target, or unreachable is pushed down no matter how good its keywords look.

RUNS UNDER: ✓ Offline | ✓ CPU-only | ✓ No GPU | ✓ Deterministic | ✓ bm25s backend

HOW IT READS A CANDIDATE

28 features separate keyword lists from recruiter-plausible fit.

Each signal is computed deterministically from profile facts — and every rank decomposes back into them.

Skills & depth
8 features

core / nice-to-have match, skill depth, endorsement trust, assessment scores, GitHub

Career evidence
8 features

product-company ratio, IR/ranking experience, production evidence, seniority, trajectory, scale

Experience fit
4 features

years-of-experience band, education, ML/AI tenure, open-source signal

Behavior
6 features

availability, engagement, responsiveness, interview reliability, profile quality, notice

Logistics
2 features

preferred-location match, willingness to relocate

Multiplicative guardrails

Behavioral, honeypot & disqualifier multipliers gate the base score so traps can’t buy their way to the top.

SEE IT WORK

The shortlist is real: every rank is traceable.

Live output from the released 100K pool, plus the exact feature values and guardrails behind the #1 candidate.

Ranked output

submission.csv top 6

Rank	Candidate evidence	YOE	Score
1	**Senior ML Engineer** - Zomato hybrid retrieval + ranking at 50M+ queries	7.2	1.000
2	**Senior NLP Engineer** - Niramai semantic search + embeddings + strong response	7.8	0.940
3	**Senior ML Engineer** - Genpact AI IR, pgvector, Sentence Transformers	6.1	0.933
4	**Senior AI Engineer** - Apple FAISS + fine-tuning + 30-day notice	5.9	0.931
5	**Senior ML Engineer** - Flipkart OpenSearch + LLM fine-tuning; logistics flag	8.1	0.921
6	**Senior AI Engineer** - Netflix Elasticsearch + Qdrant + PEFT	7.8	0.902

Readout: the top six are senior AI/ML/NLP/search profiles with production evidence; tie-breaks are deterministic by candidate_id.

Why #1 wins

CAND_0018499

Senior ML Engineer at Zomato | 7.2 yrs | Noida | 15-day notice

Feature	Value
core_skill_match	0.982
production_evidence	1.000
ir_ranking_experience	1.000
product_company_ratio	1.000
career_trajectory_score	1.000

behavior x1.10 | honeypot x1.00 | DQ x1.00

Grounded reason: *shipped retrieval/ranking; skills include Learning to Rank + Embeddings; response 0.61.*

WHY IT HOLDS UP

Stuffers and honeypots never reach the shortlist.

Measured on the released pool — the guardrails do real work, not decoration.

Keyword stuffers buried

570+

Lowest rank a non-target-title profile with a high AI-skill match reaches. None enter the top 100.

Honeypots blocked

53 → 0

Impossible profiles detected, then zero emitted in the top 100. Challenge DQ threshold is >10%.

Unavailable down-weighted

×0.42

Behavioral multiplier on an 8-yr profile with 12% response & not-open-to-work — pushed off the shortlist.

Plain-language fit is captured too: skill matching runs over the full profile text, not just the skills array — so an engineer who *describes* building a recommender (without the buzzwords) still scores on core fit.

VALIDATION

An independent LLM judge confirms the top-10 — it isn't self-graded.

We avoided grading our ranker with its own heuristics. A separate judge scored a stratified sample.

0.894	0.873	0.912	1.00
NDCG@10	NDCG@50	MAP	P@10

How it was judged

249 stratified candidates — ranker top ranks + higher heuristic tiers + a random control

Judge gemini-2.5-flash scores each 0–5 vs the JD rubric: read careers, down-weight unavailable

Top-10 tiers all 4–5 by the independent judge

Not propped up by our own labels

The judge composite (0.896) is slightly higher than our non-circular heuristic harness (0.881) — the ranking is strong by an outside opinion, not just our own.

Honest framing: this is a development-time proxy on a 249-candidate sample, not the hidden challenge score. It tells us the top of the list is real — it is not a leaderboard claim.

TRADEOFFS & LIMITS

Reproducibility beats raw model power.

The deck does not hide limitations; it turns them into Stage-5 defense points judges can verify.

Defensible decisions

shipped path

- **No LLM per candidate**
  100K hosted calls break the CPU/no-network budget and make reproduction fragile.
- **Dense embeddings tested, then kept off**
  NDCG@10 was flat while runtime and complexity rose, so the simpler ranker won.
- **Guardrails are multiplicative**
  Impossible, off-target, or unreachable profiles cannot buy rank with keywords.
- **Deterministic reasoning**
  Same input gives the same CSV; every sentence points to real profile facts.

Next hardening moves

honest roadmap

- **Hidden ground truth**
  The official leaderboard is unseen, so proxy eval is evidence - not a score claim.
- **Conservative hard-trap recall**
  53 hard traps are blocked; softer suspicious profiles are down-weighted, not over-zeroed.
- **Learned ranking layer**
  Future labels could train LambdaMART/XGBoost over these same features.
- **Stage-3 fallback**
  If multiprocessing is constrained, --workers 1 preserves the exact same ranking output.

REPRODUCIBILITY

One command reproduces the shortlist — offline, well under the five-minute limit.

Built for the Stage-3 sandbox: it runs unmodified inside the challenge's compute box.

```
>_ python rank.py --candidates ./candidates.jsonl --out ./submission.csv
```

CONSTRAINT	CHALLENGE LIMIT	MEASURED
Runtime	≤ 5 min (300s)	**184 s** ✓
Memory	≤ 16 GB RAM	**4.33 GB peak** ✓
Compute	CPU only	**CPU only** ✓
Network	off during ranking	**fully offline** ✓

Submission package

- ✓ GitHub repo — clean, tested, reproducible
- ✓ submission.csv — top-100 + grounded reasoning
- ✓ METHODOLOGY.md — every choice, with the why
- ✓ Live sandbox — HuggingFace Space
- ✓ Independent eval evidence committed (249 labels)

THE TAKEAWAY

We don’t rank resumes.
We rank hireability.

- Reads careers & behavior, not buzzwords
- Every rank explained by 28 named features
- Top-10 verified by an independent judge
- Blocks honeypots, stuffers & the unavailable
- 100K scored offline, on CPU, in 184s